INTRO

OPS Roadmap

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SECTION I

Meet the Officers!

OPS Leadership



Marvin Nguyen
OPS Lead Instructor



Abishek Vinujudson
OPS Lead Instructor



Anika Saha
OPS Lab Supervisor

OPS Instructors

- All your instructors are at the front!
- Everyone stand up and introduce yourself :)

SECTION II

Icebreaker

Talk to your Neighbors!

- **Name, year, major, fun fact**
- Some more ideas for conversation:
  - Where is your dream travel destination?
  - Where are you originally from?
  - Favorite cuisine
  - Least favorite school subject

SECTION III

IEEE Overview

IEEE
Student Branch at UC Irvine

IEEE@UCI

Who are we?

We are a student branch of the global IEEE, a professional organization with a mission to advance technology for humanity.

Where?

- Our lab is in ICS 225

What do we offer?

- Technical Workshops
- Professional Networking
- Micromouse
- Open Project Space

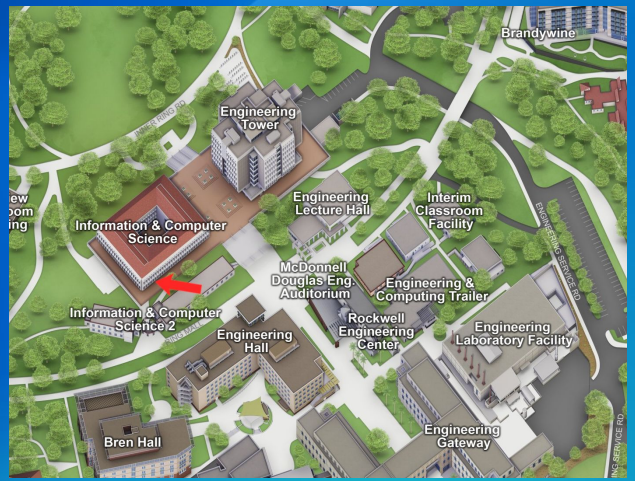




The IEEE Room

- **Where?**
  - ICS 225
  - Giant Post-it “IEEE” on the window
  - You can also find us on Google Maps as “IEEE Student Branch at UCI”
- **When is it open?**
  - Check # 🏠 the-lab
- **What do we have?**
  - 3D printers
  - Soldering Stations
  - Textbook Library
  - Study space
  - Lab Equipment





Click here for directions

IEEE
Student Branch at UC Irvine

Fall 2026 Events

Technical Workshops:

- PCB Buck Converter (Week 4)
- ESP32 and Breakout Board (Week 6)
- Full H-bridge Motor Driver (Week 6)
- Signal Generator Soldering (Week 7)
- Data Extraction from Receipts using AI (TBD)

Professional Development:

- Medtronic Event 10/14 4-5 PM
- RTX Info Session 10/28 12-1 PM

General Meetings - Tuesdays, 7-8 PM:

- Odd Weeks: GMs
- Even Weeks: IFamilieee

SECTION IV

OPS Course Overview

Lectures and Workshops

- **Lectures**
  - **Hosted every other week (odd weeks)**
  - Introduce topics that will be reinforced by the projects
  - Recorded and posted along with slides on the website
- **Workshops**
  - **Hosted every other week (even weeks)**
  - Workshops are **interactive activities** to help complete part of the project
  - First workshops is next week, and covers soldering

Lab Sessions

- Get **one-on-one support** to complete projects
- Weekly lab sessions at the IEEE Room (ICS 225)
    - Hosted by **Lab Supervisor** and **Lab Instructors**
    - **Start Week 3 (this week!)**
    - Hours posted on Canvas
- **No appointment required**



Canvas

- **Weekly Announcements**
  - Upcoming Projects, Lectures, and Workshops
  - Important Deadlines
- **Assignment Submissions**
  - Deliverables for projects are described on the assignment Canvas page as well as on the project specifications page
- **Lecture, Workshop, and Lab Schedule**
  - See the entire schedule for the active Quarter

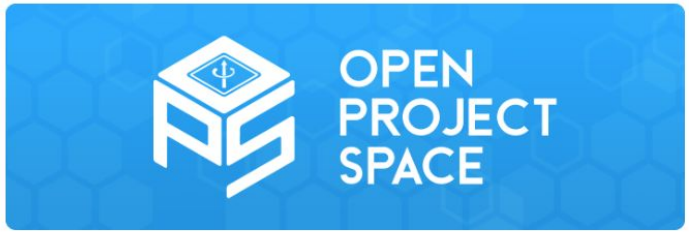


Canvas

2024-25 Academic Year

- Home
- Announcements
- Assignments
- Modules
- Syllabus
- People

Open Project Space - 2024-2025



Syllabus **Schedule** **Modules**

Contact Us

Name	Email	Discord Tag
Lead Instructor		

- View Course Stream
- View Course Calendar
- View Course Notifications

To Do

- Welcome to Open Project Space!
  Open Project Space - 2024-2025
  Oct 16 at 8:32am
- Project 1: LED There Be Light
  Open Project Space - 2024-2025
  1 points |
  Nov 1 at 11:59pm
- Project 2: 555 Piano
  Open Project Space - 2024-2025
  1 points |
  Nov 15 at 11:59pm
- Project 3: RGB LED Wizard
  Open Project Space - 2024-2025
  1 points |
  Nov 29 at 11:59pm
- Project 4: SunDial
  Open Project Space - 2024-2025
  1 points |
  Dec 13 at 11:59pm

Discord

- **General Channel**
  - #ops-general
  - Channel where you can chat and ask general questions
  - Feel free to introduce yourself here!



- **Project-Specific Questions**
  - #ops-help
  - Create posts and ask questions for project help here!
  - Include images, code snippets, and whatever else to thoroughly explain the issue

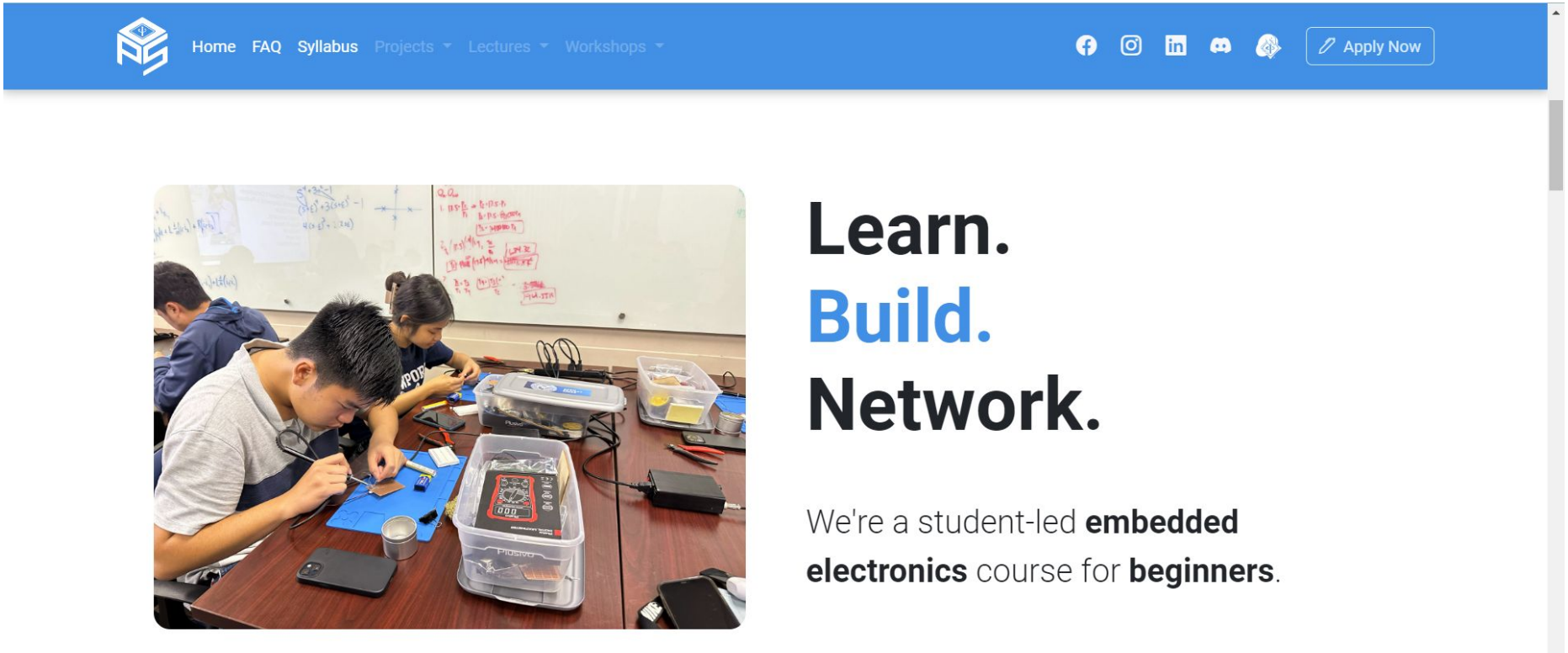
Website

- **Visit the website at ieee.ics.uci.edu/ops (bookmark it now!)**

Overview:

- **Course Syllabus**
  - You can use this as an overview of the full curriculum
- **Lectures and Workshops**
  - The slides and video recordings of lectures and workshops (when available) are posted on the site
- **Project Specifications**
  - Each project specification is made available at the start of the next assignment
  - The pages include links and hoverables to aid your project work

Website



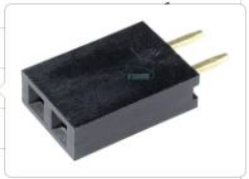
[Visit the Website]

What's in my Kit?

- **Parts**
  - Refer to the syllabus for the **full list of parts** in your kit
    - Use the project specifications page to look for parts (hover your mouse over a part)
    - There may be more parts than the quantity listed in the syllabus
    - Missing a part, or a parts no longer working? Please visit the IEEE Room for a replacement
- **Nametag**
  - Flip the lid of your kit, and **write your contact info** down (in case you misplace the kit)

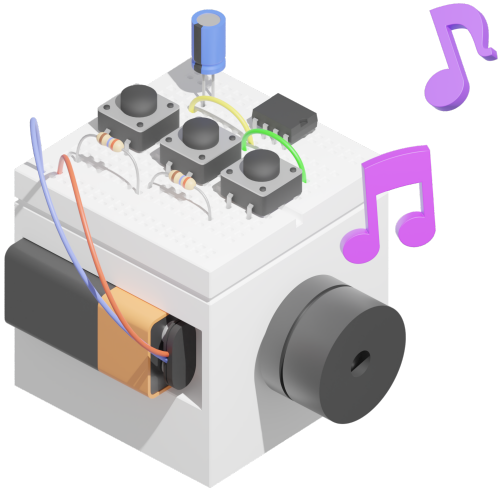
Parts

Part Name	Qty
Jumper Wire	?
Breadboard	1
Battery, 9V	
9V Snap Connector	
Header, 2.54mm, Female, 1x2	
LED, 2V	
Slide Switch	
Resistor, ?Ω	1
Perfboard	1



Projects

- Projects are **released the same day as the lecture** (every two weeks)
- Deadlines are the night of the next lecture (11:59PM)
    - **Deadlines are recommendations** to keep pace with the course
- All projects within a quarter are due at the end of the quarter
    - (end of finals week)
- Your OPS membership is contingent upon the completion of Project 1



SECTION V

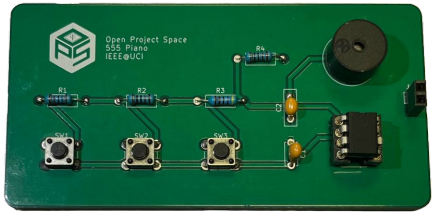
OPS Roadmap

Fall Quarter Projects

- **Project 1: LED there be Light**
  - Build an LED circuit with a switch and solder it to a perfboard.



- **Project 2: 555 Piano**
  - Create an electronic piano with the 555 Timer IC, and solder it to a printed circuit board.



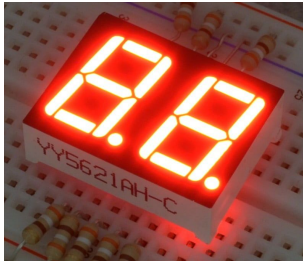
- **Project 3: RGB LED Wizard**
  - Build and program a dimmable RGB LED using the ESP32 and potentiometers.

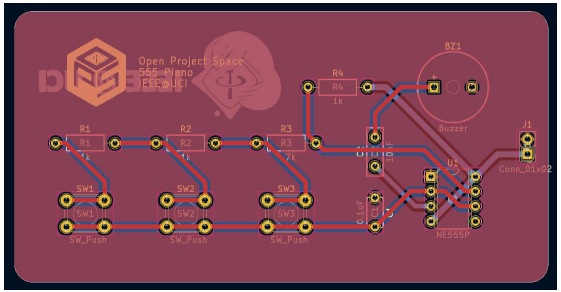
- **Project 4: Sonar**
  - Build a sonar that detects nearby objects using a servo motor and ultrasonic sensor



Winter Quarter Projects

- **Project 5: IPoduino**
  - Build an mp3 player and learn basic hardware communication.
- **Project 6: Weather Station**
  - Build a weather station that wirelessly transmits temperature and humidity data to an indoor display.
- **Project 7: Kitchen Timer**
  - Create a three-button kitchen timer using software interrupts and timers
- **Project 8: PCB Design With KiCad**
  - Design a PCB for the 555 Blinking LED or Piano.

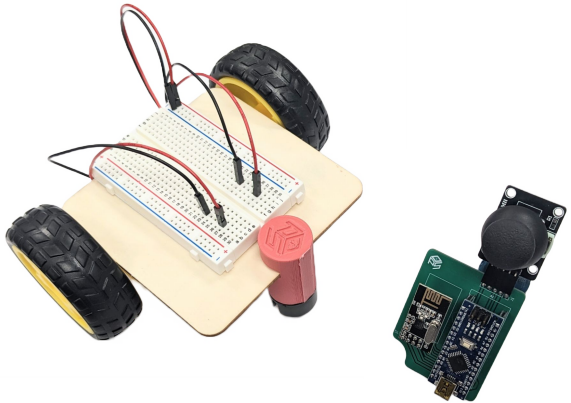




Spring Quarter Projects

- **Capstone RC Rover**
  - Build and control an autonomous rover remotely with a custom PCB(s).
  - Any questions on any project?





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